Keywords: computer-use agents • environment synthesis • synthetic training data • GUI agents

Agent World Model in the Wild: Computer-Use Agent Environment Synthesis

Persona-Grounded WildAWM Synthesises 341 Long-Horizon Computer-Use Environments; Qwen Models Trained There Generalise to 8 OOD Benchmarks

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arXiv:2610.26828 • October 2026

Training general-purpose computer-use agents requires controllable environments with verifiable ground truth—a resource that is scarce, expensive to build manually, and impossible to extract from screen recordings without execution-based verification. WildAWM synthesises such environments by grounding them in synthetic user personas: it generates executable desktop worlds with persona-appropriate websites, MCP services, and files, then synthesises long-horizon cross-application tasks and filters them through execution-based quality gates combining heterogeneous agent rollouts, cross-judge consistency, and sycophancy-probing counterfactuals. The resulting 341 environment-task pairs, which GPT-6-Astra solves at only 60%, transfer well: Qwen models trained exclusively on them gain substantially across 8 out-of-distribution computer-use benchmarks.

AT A GLANCE

Problem: Realistic, verifiable computer-use training environments are scarce; scripted generators produce tasks too simple to transfer.

Why it matters: Frontier models are approaching human performance on curated benchmarks; progress now requires harder, more diverse training data.

Method: Persona-grounded environment synthesis with three-tier quality gate: heterogeneous rollouts, cross-judge consistency, and counterfactual sycophancy probes.

Result: 341 long-horizon pairs; GPT-6-Astra at 60%; Qwen models trained on synthetic data generalise to 8 OOD benchmarks.

Evidence: OOD evaluation on web, desktop, and computer-use tasks; quality gate ablation confirms each tier's contribution.

The Big Picture

The ceiling for computer-use agents is no longer model architecture or training objective—it is training data. Frontier models like GPT-6-Astra now score above 90% on the original OSWorld and ScreenSpot benchmarks; further gains require harder, more diverse, and more verifiable training environments.

Building such environments by hand does not scale. WildAWM takes a generative approach: start from a synthetic user persona (a profile describing the person's job, preferences, and daily computer activities), derive an executable personal-computer environment from that persona, and synthesise tasks that are grounded in and verifiable against that world.

The result is a data pipeline that can, in principle, generate environments indefinitely—limited only by the diversity of the persona generator and the quality of the downstream verifier.

Why Existing Approaches Fall Short

Hand-curated benchmarks (OSWorld, ScreenSpot, AndroidWorld): Expensive to build; fixed task distribution that saturates as models improve; evaluation sets cannot scale to training-set sizes.

Rule-based environment generators: Tasks are formulaic ("click button X on website Y"); they do not resemble the cross-application, multi-step workflows that real users perform.

Web-crawled demonstrations: Ground-truth completion is difficult to verify; rewards are noisy; task length and difficulty cannot be controlled.

Simulation replay: Requires access to the original software environment; does not generalise to novel configurations.

Core Mechanism

Persona-grounded environment synthesis: An LLM generates synthetic user personas covering profession, daily activities, and software usage patterns. From each persona, WildAWM constructs an executable desktop environment: a browser populated with persona-relevant web pages, MCP (Model Context Protocol) services simulating the APIs the persona would use, and a local file system with persona-appropriate documents.

Cross-agent interface (XAI) task synthesis: Tasks that require actions across multiple applications (e.g., "find the flight mentioned in your last email, book it, and create a packing list in your documents") are synthesised as long-horizon plans with verifiable sub-goals. These XAI tasks mirror the cross-application coordination that distinguishes real user workflows from benchmark tasks.

Three-tier quality gate:

1. **Heterogeneous rollouts:** Multiple agent architectures attempt each task. Tasks trivially solved by all (too easy) or unsolvable by any (unverifiable) are filtered.
2. **Cross-judge consistency:** Multiple judge models evaluate completion; pairs with inconsistent verdicts are rejected.
3. **Counterfactual sycophancy probes:** Judges are presented with deliberately incorrect outcome feedback; judges that agree with false outcomes are excluded.

Technical Formulation

Let $u \sim P_{\text{persona}}$ be a synthetic user persona, $e \sim P_{\text{env}}(\cdot \mid u)$ the derived environment, and $\tau \sim P_{\text{task}}(\cdot \mid e)$ a synthesised task.

The quality gate passes (e, τ) if:

$$Q(e,\tau) = \mathbf{1}[\text{ConsistencyScore}(e,\tau) > \delta_c] \cdot \mathbf{1}[\text{RobustnessScore}(e,\tau) > \theta_\tau]$$

The *robustness score* exploits LLM sycophancy to stress-test verifiers:

$$\text{RobustnessScore}(e,\tau) = 1 - \frac{1}{|C|} \sum_{c \in C} \mathbf{1}[\text{Judge}(e,\tau,c_{\text{false}}) = \texttt{correct}]$$

where c_{false} is a deliberately incorrect outcome label; a judge that accepts it is flagged as sycophantic and excluded from the verifier pool.

A Qwen model π_θ is then trained with behavioural cloning on the curated pairs followed by RL with the verifier as the reward model:

$$\mathcal{J}(\theta) = \mathbb{E}_{(e,\tau)\sim\mathcal{D}_{\text{curated}}}\left[\mathbb{E}_{a\sim\pi_\theta(\cdot|e,\tau)}[V(e,\tau,a)]\right]$$

where V is the quality-gated verifier score.

Learning or Inference Procedure

1. Sample personas; generate environments via LLM; synthesise XAI tasks.
2. Apply the three-tier quality gate; retain pairs that pass all checks.
3. Train Qwen models on curated pairs with behavioural cloning, then RL with the verified reward.
4. Evaluate on held-out OOD benchmarks: no fine-tuning on any target benchmark.

What the Guarantee Says

The counterfactual sycophancy probe provides a robustness guarantee over the verifier pool: any judge that accepts a known-false outcome is excluded, bounding the false-positive rate of the verification signal. The consistency check across heterogeneous judges bounds the variance of the reward signal. Together, the two gates ensure that the training signal is both accurate and stable, which is the precondition for effective RL in the synthetic environment.

Experimental Findings

- **Dataset:** 341 long-horizon task-environment pairs, each requiring up to 100+ agent steps. GPT-6-Astra solves only 60%, confirming non-trivial difficulty.
- **OOD generalisation:** Qwen models trained exclusively on WildAWM environments gain substantially across 8 OOD benchmarks spanning web browsing, email management, file management, coding assistance, and general computer-use tasks.
- **Cross-application advantage:** Gains are largest on multi-application coordination tasks, the XAI task category WildAWM specialises in.
- **Quality gate ablation:** Removing any single tier of the gate degrades training-set quality (measured by human evaluation) and downstream performance.

Ablations and Interpretation

Without persona grounding (random environments): Task-environment pairs are less realistic; downstream benchmark performance drops, especially on tasks requiring knowledge of realistic file and browser content.

Without counterfactual probes: Some judges are sycophantic and accept false completions; reward signal becomes noisier; downstream performance degrades.

Without cross-judge consistency: High-variance verification labels slow convergence and reduce final performance.

Why synthetic environments transfer: Persona-grounded synthesis produces task distributions closely aligned with real user workflows; the XAI structure requires the same cross-application reasoning as real tasks, rather than isolated widget interactions that do not compose.

Reference

Zhaoyang Wang, Qianhui Wu, Huaxiu Yao, et al. **Agent World Model in the Wild: Computer-Use Agent Environment Synthesis**. arXiv:2610.26828, October 2026. https://arxiv.org/abs/2610.26828